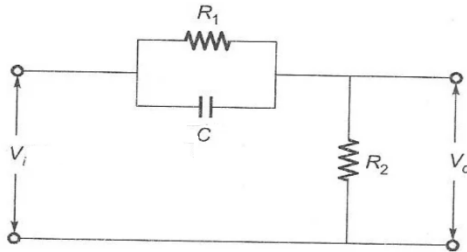


Answer all questions. Write the answers in the space provided.

1. Determine the input voltage $v_i(t)$ if the steady state response of the circuit shown in figure is $v_o(t) = 4 + 5\sin(2t + 32^\circ)$. $R_1 = 1\Omega$, $C = 0.5F$, $R_2 = 1\Omega$ [6]

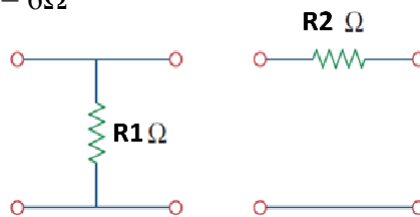


$$8 + 10\sin(2t + 13.565^\circ)$$

2. The transfer function of a network is $G(s) = \frac{s}{(s-2)(s^2+2s+5)}$. What is the phase of the network at $\omega = 6$ rad/sec. [3]

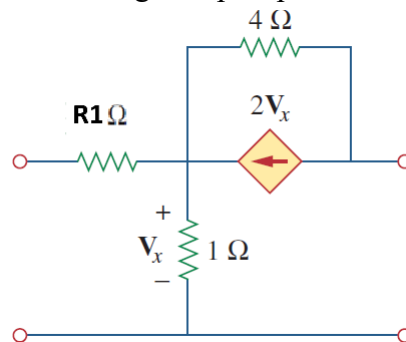
$$-177.2737^\circ$$

3. If the two two-ports networks shown below are connected in cascade, then what is the parameter D? $R_1 = R_2 = 6\Omega$ [3]



$$2$$

4. Find Y_{12} of the following two-port parameters. $R_1 = 3\Omega$ [5]

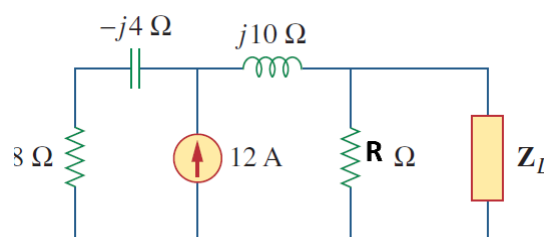


$$0.2$$

5. The two-wattmeter method is applied to a three-phase three-wire, 90 V ABC system with the meters in the lines B and C, $W_B = 836$ W and $W_C = 224$ W, Find the impedance of the balanced delta-connected load. [6]

$$18 \angle 45^\circ$$

6. Find the load impedance Z_L that absorbs maximum power transfer. $R = 2\Omega$. [2]



$$1.71 - j0.1765$$